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MILITARY SPECIFICATION

ENGINES, AIRCRAFT, RECIPROCATING, MODEL SPECIFICATION FOR
(OUTLINE AND INSTRUCTIONS FOR PREPARATION)

This specification has been approved by the Department of the Air Force and by the Navy Bureau of Aeronautics.

01. SCOPE

01.1 This specification establishes the form to be used by manufacturers in the preparation of specifications for reciprocating aircraft engine models.

02. APPLICABLE DOCUMENTS

02.1 The following specification, of the issue in effect on date of invitation for proposals, forms a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-E-25109	Engines, Aircraft, Reciprocating, General Specification for

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

03. APPLICATION

03.1 A complete engine specification conforming to the outline and instructions for preparation contained in the "model specification" and as specified herein, shall be prepared for each specific engine model. New engine model designations will be assigned by the Government. When a new model designation is assigned, a new engine model specification shall normally be prepared with a new specification number assigned thereto by the engine manufacturer. It is desired that no new engine model designation be implemented by appendixes or variants to an existing model specification, and in no case shall amendments be used for this purpose. It is desired that no changes to an engine model specification shall be submitted to the using service by means of amendments or revision pages

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prior to approval of the engine model specification and its incorporation in a contract. Revisions, by amendment form, to an approved engine model specification which forms a part of a contract will be acceptable to the using service. Each amendment shall be approved and shall include and supersede the previous amendment.

03.2 The headings and numbering of sections and paragraphs of the model specification correspond to those of Specification MIL-E-25109. Those paragraphs which need not be included in the model specification have been omitted from the outline. Omission of reference in the model specification to a particular requirement of Specification MIL-E-25109 shall indicate complete compliance therewith. When departures from the requirements of this specification and of Specification MIL-E-25109 are necessary, the details of such departures shall be stated as specific requirements bearing the same section, paragraph heading, and numbering as in Specification MIL-E-25109.

03.3 Parenthetical sentences, phrases, and words are included herein for the guidance of the engine manufacturer in connection with the preparation of the model specification. Parenthetical statements need not be copied in the model specification.

03.4 The specification number shall be the number assigned by the engine manufacturer. When revisions are made, they shall be designated by the use of a dash and a letter following the number, with a revision date therefor, which shall be shown on page 1 only. Only the specification number, and the revision suffix letter if applicable, shall be shown on subsequent pages.

03.5 For purposes of permitting preliminary evaluation of a proposed engine design, or for release of approved engine performance characteristics in connection with an airplane design competition, the engine manufacturer may submit a preliminary model specification. However, a complete approved model specification will be required for a contract.

04. MODEL SPECIFICATION

04.1 The format and instructions for the preparation of the model specification follow (see Sections 1 through 4).

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(Number and title.- The number and title shall be shown as follows:)

(Spec No.)_____
(Date)_____
(a) Revised_____(Date)
(b) Revised_____(Date)

MODEL SPECIFICATION

ENGINE, AIRCRAFT, RECIPROCATING: _____

(Insert Service type and model designation if assigned)

(NAME OF CONTRACTOR)
(CONTRACTOR'S MODEL DESIGNATION)

1. SCOPE

1.1 Scope.- This specification covers the requirements for the _____ _____(insert the Service type and model designation) engine.

1.2 Classification.- The _____(insert type and model) engine is a _____ (insert briefly the description of the salient features of this model engine).

2. APPLICABLE DOCUMENTS

2.1 The following specification forms a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-E-25109	Engines, Aircraft, Reciprocating, General Specification for

2.2 The following publication shall be applicable to this specification, except as specified herein:

PUBLICATIONS

Air Force-Navy Aeronautical Bulletin

No. 343	_____(Insert revision and amendment designation) dated _____(insert date) entitled "Specifications and Standards Applicable to Aircraft Engines and Propellers, Use of"

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3. REQUIREMENTS

3.3 Performance characteristics.- The ratings and curves shown are based on the terms and standard conditions defined in Specification MIL-E-25110.

3.3.1 Ratings.- The performance ratings shall be as listed in table I. These data are based on the use of fuel conforming to Specification MIL-F-5572, grade _____, with no loading of accessory drives, and oil conforming to Specification MIL-L-6082, grade _____. With water-alcohol injection, the alcohol conforms to Specification O-M-232, grade A, or to Specification MIL-A-6091.

TABLE I

Ratings and specific fuel consumption

Ratings	BHP	RPM	Alt feet	SFC	Max allowable temperature			Max static exhaust pressure (In. Hg Abs)	Min dry static air press. at carb. inlet (In. Hg Abs)
					Head (°F)	Base (°F)	Cool (°F)		
Takeoff									
Military									
(a) Sea level									
(b) 1st crit.									
(c) 2nd crit.									
(d) 3rd crit.									
Normal									
(a) Sea level									
(b) 1st crit.									
(c) 2nd crit.									
(d) 3rd crit.									
Dive speed									

NOTES: (In addition, specific fuel consumption for 90, 80, and 70 percent normal sea level rated power on a normal rated power and speed propeller load shall be given.)

(The maximum static exhaust back pressure and the minimum dry static air pressure at the carburetor inlet shall be given only for engines intended for use with turbosuperchargers.)

(For purposes of conducting the Qualification tests, and when an approved carburetor setting is not established, values of specific fuel consumption on normal rated power and speed propeller load shall be supplied for each specified simulated altitude test condition and for the condition of 100 percent normal rated power and 110 percent normal rated speed.)

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(3.3.1.1 Maximum power on normal rated power and speed propeller load to which operation with best economy mixture strength is permissible shall be stipulated. The minimum power and speed for best economy mixture strength shall also be stipulated. All points between these two points shall be permissible for operation at maximum economy.)

(3.3.1.2 For multistage gear-driven supercharged engines requiring external intercooling between stages, the ratings given above shall be based on 1.5 inches of mercury total intercooler and duct pressure loss for auxiliary stage ratings; and the powers designated shall be those obtainable with the auxiliary stage regulator in operation.)

(3.3.1.3 For multistage engines of all types requiring external intercooling between stages, the ratings as given above shall be based on a minimum of 100°F(37.8°C) carburetor air temperature; or, if a higher value is specified, the performance shall be based upon the specified higher value.)

3.3.2 Estimates.- (The following instructions shall be observed in the preparation of curves and data specified herein.)

(3.3.2.1 If the engine manufacturer recommends a mixture heater (hot spot) for aircraft installation, all of the data shall represent operation with the heater in use. A statement shall be included on each curve sheet as to whether or not a heater is used. If the heat is obtained from the exhaust, the number of cylinders connected shall be specified.)

3.3.2.2 Performance curves.- (The following performance curves shall be furnished as part of this specification. These curves shall illustrate performance attainable with each mixture control position. Separate sets of curves shall be supplied for each supercharger gear ratio. In the case of engines using uncommon arrangements or having uncommon characteristics, curves adequately illustrating powers and specific fuel consumptions attainable at various speeds and altitudes over the permissible operating range of the engine shall be supplied. For engines furnished with boost or power controls, or auxiliary stage supercharger regulators, the curves shall illustrate the powers obtainable with these controls or regulators in operation.)

3.3.2.2.1 Horsepower versus altitude.- (These curves shall show horsepower versus altitude at speeds from 40 percent normal rated speed to military rated speed inclusive. The curves shall illustrate powers corresponding to the engine ratings (or corresponding to the maximum recommended bmep) from sea level (or minimum recommended shift altitude) to the critical altitude for the power-rpm combination illustrated. Full throttle performance shall be illustrated from the critical altitudes to the guaranteed altitude of the engine. If the engine is equipped with boost control or regulator, the curves shall illustrate performance attainable therewith. Manifold pressure requirements shall be superimposed as dotted lines, except that these data may be eliminated from preliminary bid specifications.)

3.3.2.2.2 Specific fuel consumption versus engine speed.- (These curves shall show specific fuel consumption versus engine speeds at constant powers. These curves shall include a range of powers from 20 percent normal rated to military rated power, and from 40 percent normal rated speed (or the speed corresponding to the maximum recommended bmep for the power illustrated) to military rated speed. The specific fuel consumption will be assumed to be constant regardless of altitude unless information to the contrary is supplied on each curve sheet. Points of maximum recommended bmep shall be indicated on each horsepower curve but shall not influence the Qualification test procedure.)

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(3.3.2.2.3 In the case of engines intended for use with exhaust-driven superchargers, the curves showing horsepower versus altitude and specific fuel consumption versus engine speed shall illustrate performance attainable without the aid of the exhaust-driven supercharger, and with standard atmospheric conditions prevailing at the carburetor inlet and exhaust ports. In addition, a curve similar to that for horsepower versus altitude shall be supplied illustrating power and airflow versus carburetor air pressure altitude (static) and the stipulated exhaust pressures. Fuel consumption curves similar to those for specific fuel consumption versus engine speed shall be supplied illustrating fuel consumptions attainable at the stipulated exhaust back pressures. These data shall be based on values of carburetor air temperature specified under table I.)

(3.3.2.2.4 Airflow and temperature rise through the auxiliary stage corresponding to each rating shall be provided for geared auxiliary stage engines requiring external intercooling.)

3.3.3 Specific oil consumption.- The specific oil consumption shall not exceed _____ pounds per bhp-hour at military power and speed and _____ pounds per bhp-hours at normal rated power and speed and _____ pounds per bhp-hour at 70 percent normal rated power and 89 percent normal rated speed with the lowest degree of supercharging used at sea level.

3.3.4 Altitude operating limits.- The complete engine shall function satisfactorily up to and including an altitude of _____ feet with the following limiting operating conditions.

3.3.4.1 (The following paragraph is to be inserted verbatim in each model specification.):

> "Following flight tests conducted by the aircraft manufacturers, sufficient data shall be submitted to the engine manufacturer to enable him to check the performance of the engine air scoops and exhaust systems on engine performance and to submit his recommendations to the using service. If changes are required, they shall be subject to contract negotiations."

3.3.5.2 Airspeed and altitude.- The engine shall operate satisfactorily within the ambient air temperature limits specified in Specification MIL-E-25109 throughout the following airspeed and altitude limits:

(a) Airspeed: _____ to _____ knots,
(b) Altitude: _____ to _____ feet.

3.3.6 Engine starting.- Provision shall be made for starting the engine with (designate means for meeting starting requirements).

3.3.8 Acceleration.- The complete engine shall accelerate from 50 percent normal rated speed to takeoff speed in _____ seconds when operated on a test stand with flight-type propeller set for takeoff propeller load.

3.3.9 Operating data.-

3.3.9.1 Cylinder head temperatures (air-cooled engines).- The maximum allowable cylinder head temperatures, measured at the place shown on the installation drawing, shall not exceed the following values:

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_____	°F	at takeoff power - 5 minutes operation.
_____	°F	at normal rated power to 70 percent normal rated power - continuous operation.
_____	°F	at 70 percent normal rated power and below - continuous operation.
_____	°F	at military rated power.

3.3.9.2 Cylinder barrel temperatures (aircooled engines).- The maximum allowable cylinder barrel temperatures as measured on the downstream side of the cylinder in the fillet where the cylinder barrel joins with the cylinder holddown flange, or as otherwise specified, shall not exceed the following values:

_____	°F	at takeoff power - 5 minutes operation.
_____	°F	at normal rated power to 70 percent normal rated power - continuous operation.
_____	°F	at 70 percent normal rated power and below - continuous operation.
_____	°F	at military rated power.

3.3.9.3 Coolant flow and heat rejection.- The following data shall apply to operation with coolant as specified under 3.27 entitled "Coolant temperature (liquid-cooled engines)."

3.3.9.3.1 Normal rated power and speed.- When operating with coolant outlet temperature of _____°F, (250°F minimum) at normal rated power and speed and the specified specific fuel consumption, the coolant flow shall not exceed _____ gallons per minute and the heat rejection shall not exceed _____BTU per minute (_____horsepower) with (insert knots and air temperature) air blast on the engine when measured on a dynamometer.

3.3.9.3.2 Military rated power and speed.- When operating with a coolant outlet temperature of _____°F (250°F minimum) at military rated power and speed and the specified specific fuel consumption, the coolant flow shall not exceed _____gallons per minute and the heat rejection shall not exceed _____BTU per minute (_____horsepower) with (insert knots and air temperature) air blast on the engine when measured on a dynamometer.

3.3.9.3.3 The flow characteristics of the coolant pumps shall be as shown on pages_____(insert applicable page numbers. Manufacturers shall insert curves in accordance with the following requirements, showing by notes on each curve sheet all pertinent operating conditions. These data may be obtained on bench test.)

3.3.9.3.3.1 Curves showing the variation in coolant rate-of-flow versus pressure drop in the cooling system outside the engine between limits of 5 and _____pounds (15 pounds minimum) per square inch drop, for the operating condition specified in 3.3.9.3.1 entitled "Normal rated power and speed," and with _____pounds (25.0 pounds minimum) per square inch absolute pressure at the coolant pump inlet.

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3.3.9.3.3.2 A curve showing the variation in coolant rate-of-flow versus absolute pressure of coolant pump inlet limits of 8 and _____pounds (25.0 pounds minimum) per square inch absolute, with the engine operating at 70-percent normal rated speed, at normal rated speed, at takeoff speed, and at military speed, with a coolant outlet temperature of _____°F (250°F minimum) and a pressure drop of 5 to _____pounds (15 pounds minimum) per square inch in the cooling system outside the engine. (The curve sheet shall show the actual percentage concentration and specific gravity at a designated temperature.)

3.3.9.4 Coolant flow and heat rejection in liquid intercooler or aftercooler.- When operating with a coolant inlet temperature of _____°F with coolant specified under 3.25 entitled "Coolant temperature (liquid cooled engines)" at military rated power and altitude (highest military critical altitude) and at the specified specific fuel consumption, the coolant flow shall not exceed _____gallons per minute, and the heat rejection to the coolant shall not exceed _____BTU per minute (_____horsepower) with _____(insert knots and air temperature) air blast on the engine when measured on a dynamometer.

3.3.9.4.1 Curves showing the variation in coolant rate-of-flow versus absolute pressure at coolant pump inlet between limits of 8 and _____pounds (25.0 pounds minimum) per square inch absolute, with the engine operating at 70 percent normal rated speed, at normal rated speed, at takeoff speed, and at military speed, with a coolant outlet temperature of _____°F and at a pressure drop in the intercooler or aftercooler system external of the engine of 10 to 60 pounds per square inch. (These data may be obtained on bench test.)

(3.3.9.5 Oil flow and heat rejection.- For engines not having a military rating, data for takeoff power and speed shall be furnished in lieu of military rated power and speed.)

3.3.9.5.1 Normal rated power and speed.- When operating on a rigid test stand at normal rated power and speed, with an oil inlet temperature of 185°F for grade 1100 oil, and with the specified oil pressure of _____±2 pounds per square inch at each pump, and the specified specific fuel consumption, the oil flow shall not exceed _____pounds per minute, and the heat rejection to the oil shall not exceed _____BTU per minute (_____ horsepower). (When the grade of oil and an oil inlet temperature different from the above is specified in the model specification, the data shall be supplied for the specified conditions.)

3.3.9.5.2 Military rated power and speed.- When operating on a rigid test stand at military rated power and speed, with an oil inlet temperature of 203°F for grade 1100 oil, and with the specified oil pressure of _____±2 pounds per square inch at each pump, and the specified specific fuel consumption, the oil flow shall not exceed _____pounds per minute, and the heat rejection to the oil shall not exceed _____BTU per minute (___horsepower). (When the grade of oil and an oil inlet temperature different from the above is specified in the model specification, the data shall be supplied for the specified conditions.)

3.3.9.5.3 Liquid-cooled engines.- (For liquid-cooled engines, the data required under 3.3.9.5.1 entitled "Normal rated power and speed" and 3.3.9.5.2 entitled "Military rated power and speed" shall be furnished based on conditions as specified for liquid-cooled engines for normal rated power and speed, and military rated power and speed under 3.3.9.3 entitled "Coolant flow and heat rejection.")

3.6 Drawings and data.- The following _____(insert manufacturer's corporate name) drawings and data form a part of this specification: (designate applicable drawings as specified in Specification MIL-E-25109).

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3.10 Dry weight of complete engine.- The dry weight of the complete engine, including its components, shall not exceed _____pounds.

3.10.1 Weights of additional equipment.- (When additional equipment is furnished with the engine, such as exhaust collectors, accessories, external fan gear bearings and fan controls, carburetor air ducts, cooling fans, etc, such items shall be listed in the model specification in tabular form. These items are not to be included in the engine dry weight.)

3.13 Torquemeter.- The engine shall (not) be equipped with a torquemeter.

3.15 Helicopter engine effective flywheel requirements.- The minimum effective inertia to provide satisfactory operation shall be _____lb/in/sec^2 (if applicable).

3.16 Compression ratio.- The engine shall have a compression ratio of _____ (expressed to nearest 0.05).

3.17.1 Propeller drive.- Provision shall be incorporated for _____(designate single acting or double acting hydraulic governing system or dual rotation governing or a system without oil transfer provisions to propeller). (Designate also, provisions for two-speed reduction gear controls when provided.) The engine shall have a No. _____ (shaft number(s)) (designate single or dual) _____rotation propeller shaft(s). The direction of rotation when viewed from the antipropeller end shall be _____(clockwise or counterclockwise) (if dual rotation, designate "counterclockwise for the inboard shaft and clockwise for the outboard shaft"). The propeller drive shall be _____(designate direct drive or equipped with a reduction gear ratio of _____ or equipped with multiple speed gearing with reduction gear ratios of _____ _____).

3.18.4 Impeller gear.- The impeller gear ratio shall be _____(and _____) and the impeller(s) shall be _____(and) _____inches in diameter.

3.19 Automatic engine control.- (The operating characteristics of the automatic engine control, if used, shall be described herein. The manufacturer's name and model number shall be given.)

3.19.3 Compensation for operating fluid pressure.- The minimum allowable operating pressure at the control shall be _____pounds per square inch.

3.25 Coolant temperature (liquid-cooled engines).- The cooling liquid outlet temperature for liquid-cooled engines shall be _____°F. The coolant used shall be _____percent ethylene glycol and _____percent water by volume. The maximum inlet pressure to the cooling liquid pump on liquid-cooled engines shall be _____pounds per square inch.

3.26 Cylinder temperature measurement (air-cooled engines).- The _____(insert the type((thermocouples or resistance bulbs) shall (not) be furnished with the engine.

3.27.1 Carburetor or fuel injector.- The engine shall be equipped with _____(insert manufacturer's name and model of carburetor(s) or fuel injector(s) used). The normal operating fuel pressure shall be _____pounds per square inch and the range of allowable fuel pressure shall be _____pounds per square inch to _____pounds per square inch. A reference carburetor shall (not) be submitted for approval for this engine model(s). (If no reference carburetor is submitted, then the carburetor setting shall be specified upon which a release has been previously granted for another engine.)

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3.27.1.3.2 Altitude compensation.- For airflows corresponding to 30, 50, 80, and 100 percent of normal rated power and takeoff power under carburetor entrance conditions shown in table II below, the fuel-air ratio shall not vary from that obtained with standard sea level carburetor entrance conditions by more than the percentage specified for the tabulated conditions up to the engine critical altitude for these airflows.

TABLE II

Carburetor entrance conditions

Applicable range of specific weights (lb/cu ft)	Applicable range of temperature °F	Limits (percent)
0.10 to 0.05	To be specified by mfr	±2
0.05 to 0.034		±3
0.034 to 0.018		±4

3.27.2.1 Filter.- (State whether or not the filter is furnished with the engine. When the filter is not supplied with the engine, the engine manufacturer shall specify the filter requirements herein.)

3.27.4 Water injection system.- The engine shall be equipped with a _____ (specify type and model of water regulator) water injection system, (if applicable).

3.30.5 Scavenging system.- When using an oil or diluted oil as specified in Specification MIL-E-25109, the normal operating temperature shall be maintained between _____°F and _____°F.

3.30.6 Pressure pumps(s).- The main engine oil pressure at normal rated power and speed shall be _____pounds per square inch and the engine nose section pressure shall be _____pounds per square inch.

3.30.18 Wet-sump capacity.- The sump capacity shall be _____.

3.31.2.1 Spark plugs.-

3.31.2.1.1 Type.- The engine shall be fitted with _____(insert type and model) spark plugs. The alternate spark plug approved for use is _____(insert type and model).

3.31.2.4 Magnetos.- The engine shall be equipped with _____(insert manufacturer's name and model of magneto(s) or ignition generator used) magneto(s) (ignition generator).

3.31.2.4.4 Magneto temperature.- The maximum continuous operating temperature of the magneto shall be _____°F as measured by a thermocouple located under the coil holddown screw.

3.31.2.4.5 Low-tension coil temperature.- The maximum continuous operating temperature at the bottom of the low-tension coil, if used, shall be _____°F, as measured by a thermocouple located under the coil holddown screw.

3.31.2.7 Ignition boost.- The ignition shall be boosted during starting with _____(specify starting device, impulse coupling, or type of vibrator).

3.32 Accessory drive.- The types of each drive, the number used, and the gear ratio to the engine crankshaft, the maximum permissible torque in pound-inches for continuous operation, and the maximum permissible static torque in pound-inches (include starter), and the direction of rotation when looking at the drive on the engine, shall be as follows:

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(Insert here, in tabulated form the names of the accessory drives, the type, the number used, the ratios stated to the nearest 0.001, the continuous and static torques, the direction of rotation, and the adapters and gaskets supplied.)

4. QUALITY ASSURANCE PROVISIONS

4.1 The requirements for sampling, inspection, and tests shall be as specified in Specification MIL-E-25109.

4.2 Qualification tests.- Qualification of the _____(insert type and model designation) engine shall be predicated on the satisfactory completion of tests in accordance with Specification MIL-E-25111 and approval of the test report by the using service.

4.3 Preliminary flight rating tests.- Establishment of a preliminary flight rating for the _____(insert type and model designation) engine shall be predicated on the satisfactory completion of tests in accordance with Specification MIL-E-25113 and approval of the test report by the using service.

4.4 Acceptance tests.- Acceptance tests shall be conducted on each production engine and shall consist only of those tests specified in Specification MIL-E-25112. The fuel to be used shall conform to Specification MIL-F-5572, grade _____.

5. PREPARATION FOR DELIVERY

5.1 The requirements for preparation for delivery shall be as specified in Specification MIL-E-25109.

6. NOTES (For reference only)

6.1 Intended use.- This specification establishes the form to be used by manufacturers in the preparation of reciprocating engine model specifications.

6.2 Symbols and definitions.-

6.2.1 Qualification tests.- Qualification tests are those tests accomplished on engines and components submitted for qualification to demonstrate the suitability of an engine model for production.

6.2.2 Preliminary flight rating tests.- The Preliminary flight rating tests are those tests conducted to demonstrate the suitability of an engine model for limited use in experimental aircraft flight testing.

6.2.3 Acceptance tests.- The Acceptance tests are those tests conducted on engines submitted for acceptance under contract to demonstrate suitable quality control, correct assembly, and performance.

6.2.4 Government.- The term "Government" as used in this specification should be interpreted to mean the procuring activity of the Department of Air Force and Navy Aeronautical Services.

6.2.5 Procuring activity.- The procuring activity is the Service which negotiates the engine contract.

6.2.6 Using service.- The using service is the Service whose model dash number has been assigned to the engine in accordance with ANA Bulletin No. 395.

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6.2.7 Normal rated speed (N.R.S.).- Normal rated speed(s) is the speed(s) designated by the contractor and accepted by the Government.

6.2.8 Takeoff speed.- Takeoff speed is the speed designated by the contractor and accepted by the Government.

6.2.9 Military rated speed.- Military rated speed is the speed designated by the contractor and accepted by the Government.

6.2.10 Normal rated power.- Normal rated power(s) is the power(s) designated by the contractor and accepted by the Government.

6.2.11 Normal sea level power.- Normal sea level power is the maximum power permissible for continuous use under standard sea level conditions and is the same as normal rated power at the lowest rated altitude.

6.2.12 Takeoff power.- Takeoff power is the power designated by the contractor and accepted by the Government.

6.2.13 Military rated power.- Military rated power(s) is the power(s) designated by the contractor and accepted by the Government. Military ratings will be for 30 minutes duration.

6.2.14 Maximum best economy cruise power.- Maximum best economy cruise power(s) is maximum power(s) on propeller load through normal rated power and speed at which operation with best economy mixture strength is permissible as designated by the contractor and accepted by the Government.

6.2.15 Normal rated altitude.- Normal rated altitude is the altitude at which normal rated power is developed at normal rated speed with full throttle under standard altitude conditions. (An engine with a two-speed or two-stage supercharger has two or more normal rated altitudes.)

6.2.16 Military rated altitude.- Military rated altitude is the altitude at which military rated power is developed at military rated speed with full throttle under standard altitude conditions. (An engine with two-speed or two-stage supercharger has two or more military rated altitudes.)

6.2.17 Manifold pressure.- Manifold pressure is the absolute pressure in inches of mercury measured at the point specified by the accepted installation drawing of the engine. All test values will be on a basis of dry absolute manifold pressure which is obtained by subtracting the vapor pressure from the observed absolute manifold pressure. (All manifold pressures will be determined on test by a mercury barometer alone or with mercury manometer.)

6.2.17.1 In the case of an altitude engine, a manifold pressure higher than normal rated manifold pressure will be required to develop normal rated power on the test block. This relation also applies to engines having military ratings.

6.2.18 Normal rated manifold pressure.- Normal rated manifold pressure is the absolute manifold pressure obtained with full throttle at normal rated speed and normal rated altitude under standard altitude conditions. (An engine with a two-speed or two-stage supercharger may have two or more rated manifold pressures.)

6.2.19 Military rated manifold pressure.- Military rated manifold pressure is the absolute manifold pressure obtained with full throttle at military rated speed and military rated altitude under standard altitude conditions. This manifold pressure is the maximum manifold pressure used for military flight operations, except takeoff, from sea level to military rated altitude. (An engine with a two-speed or two-stage supercharger may have two or more military rated manifold pressures.)

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6.2.20 Takeoff manifold pressure.- Takeoff manifold pressure is the absolute manifold pressure required to develop takeoff power at takeoff speed under standard sea level conditions.

6.2.21 Takeoff bmep.- Takeoff bmep is the brake mean effective pressure required to develop takeoff power at takeoff speed.

6.2.22 Sea level engine.- A sea level engine is an engine which is designed for extended periods of full throttle operation at sea level.

6.2.23 Altitude engine.- An altitude engine is an engine which is designed for extended full throttle operation at altitude and for which full throttle operation at sea level is restricted or prohibited.

6.2.24 Actual brake horsepower.- Actual brake horsepower is the actual horsepower output of the engine under the prevailing test conditions. This is sometimes referred to as "observed brake horsepower."

6.2.25 Standard conditions.- Standard conditions are the values of air temperature and pressure given in NACA Technical Note 3182. The standard humidity, for the purpose of this specification, is zero vapor pressure at all altitudes. Standard conditions will apply to carburetor or air throttle entrance pressure, temperature and humidity, and exhaust pressure.

6.2.26 Carburetor or air throttle entrance pressure.- Carburetor or air throttle entrance pressure is the static absolute air pressure at the entrance (before the screen) to the carburetor or air throttle. It will be determined on test by a mercury barometer alone or with a water or mercury manometer.

6.2.27 Specific fuel and oil consumption.- Specific fuel consumption is the weight in pounds of fuel consumed per actual brake horsepower per hour, and specific oil consumption is the weight in pounds of oil consumed per actual brake horsepower per hour.

6.2.28 Mixture adjustment.- The following definitions apply to any given speed and throttle position. "Rich" mixture adjustment is the adjustment in which the mixture control lever is in the position giving maximum fuel flow. "Best power" setting is the setting which gives the least fuel flow when maximum power is developed. "Best setting" is that setting which results in 1 percent drop in power from "best power" caused by a reduction in fuel flow. "Best economy" is that setting which results in the minimum specific fuel consumption.

6.2.29 Maximum power at any given speed and throttle position.- Maximum power is the maximum actual brake horsepower which the engine will develop at any given speed and throttle position with optimum mixture strength.

6.2.30 Propeller load horsepower.- Propeller load horsepower is the horsepower represented by a curve drawn through the point of developed horsepower at designated speed and varying directly as the cube of the speed.

6.2.31 Fuel-air ratio.- Fuel-air ratio is the ratio of fuel consumption to air consumption on a basis of weight rate-of-flow measurements.

6.2.32 Supercharger main stage.- The main stage of a multiple stage supercharger is the stage discharging mixture directly into the intake manifold system.

6.2.33 Supercharger auxiliary stage.- The auxiliary stage of a multiple stage supercharger is the stage through which the air passes before entering the main stage. Normally an intercooler and the carburetor are between the auxiliary and main stages.

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6.2.34 Normal magneto capacitance load.- Normal load will be construed to mean a capacitance equal to or greater than the capacitance of the longest magneto or ignition coil to spark plug circuit lead in the installation where the magneto or ignition coil is to be used.

6.2.35 Normal rated magneto rotor speed.- The normal rated rotor speed of a specific engine magneto will be the magneto rotor speed corresponding to normal rated engine speed.

6.2.36 Engine components.- Engine components are items of equipment furnished as parts of the engine which are required for engine operation, such as fuel metering equipment, power controls, ignition systems, etc.

6.2.37 Accessories.- Accessories are items of equipment furnished by the Government or the aircraft manufacturer which are required for aircraft operation or as auxiliaries for engine operation, such as starters, tachometer generators, aircraft hydraulic pumps, air pumps, and electrical generators.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEM PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426

REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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